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Course: Advance Machine Learning (AI)
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Cost-Aware Visual Router for Multimodal Document Retrieval

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1 Problem Statement

Existing Retrieval-Augmented Generation (RAG) systems for multimodal documents indiscriminately apply expensive vision-language retrieval (e.g., ColPali, NeMo Retriever) to all queries, resulting in prohibitive computational costs (visual retrieval consumes 20× more resources than text-based methods) and unnecessary latency for queries that could be answered through cheaper parametric knowledge or text-only retrieval, while current routing approaches rely on static heuristics that fail to account for query-specific utility of visual information and cannot dynamically escalate when initial retrieval paths prove uncertain.

2 Research Gap

- **Efficiency-Accuracy Trade-off Ignorance:** Current multimodal RAG systems lack learned routing mechanisms that optimize cost-quality trade-offs, they either use expensive visual retrieval universally or rely on brittle rule-based switches that ignore query semantics.
- **Static Knowledge Assumption:** Existing query routing methods (RouterDC, RouteLLM, GraphRouter) assume fixed LLM capabilities and fail to model how external documents dynamically shift retrieval utility across different modalities, visual retrieval may help some queries but harm others through distraction
- **Absence of Uncertainty-Aware Escalation:** No existing system combines router confidence estimation with retrieval uncertainty quantification to enable cost-efficient cascading; current cascade routing for LLMs has not been adapted for multimodal document retrieval with modality-specific cost structures
- **Missing Utility Prediction Framework:** While RAGRouter models RAG capability embeddings, it does not predict expected utility gain from visual vs. text retrieval, nor does it incorporate cost-aware escalation thresholds based on real-time uncertainty signals from retrieval quality and generation confidence
- **Benchmark Gap:** No standardized benchmarks exist for evaluating cost-efficiency trade-offs in multimodal document RAG; MTEB and FinanceBench measure accuracy but ignore latency, cost, or routing decision quality

3 Research Objectives

1. **Develop a Learned Utility Prediction Model** that estimates expected quality gain per unit cost for visual, text, parametric, and hybrid retrieval paths given a query, using multi-task learning to jointly optimize path classification and utility regression with cost-sensitive weighting
2. **Design an Uncertainty-Aware Escalation Mechanism** that combines epistemic uncertainty (router confidence), aleatoric uncertainty (retrieval score variance), and

generation uncertainty (token perplexity) to trigger cost-efficient escalation from cheap to expensive paths only when initial results are unreliable

3. **Create a Dynamic Training Data Collection Pipeline** that generates ground-truth routing labels by executing all retrieval paths offline and computing utility scores (answer quality / computational cost), enabling supervised learning of optimal routing policies without human annotation

4. **Implement and Validate a Production-Ready Visual Router** for ColPali/NeMo-based document RAG that achieves $\geq$85% accuracy of always-using-visual-retrieval while reducing average cost by $\geq$40%, with automatic escalation ensuring $\geq$5% accuracy degradation on complex visual queries

5. **Establish Evaluation Protocols and Benchmarks** for cost-aware multimodal RAG routing, including metrics for routing accuracy, cost-efficiency curves, and escalation effectiveness, filling the gap in standardized efficiency benchmarks for document understanding systems

4 Proposed Methodology Diagram

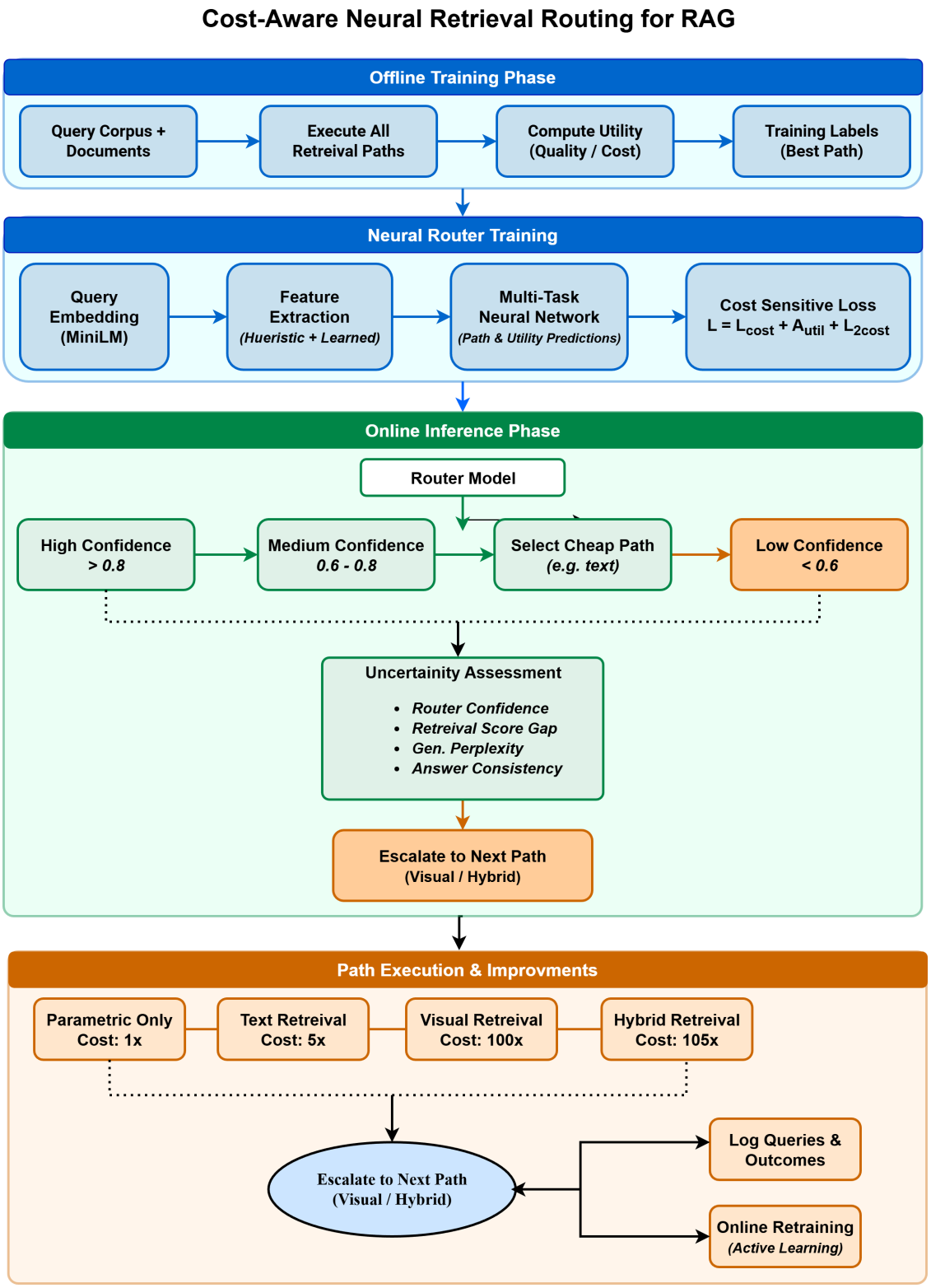

Figure 1: Cost-Aware Neural Retrieval Routing for RAG